Algebra and Analysis and Topology yay

Raymond Xu

12 oktober 2026

Inhoudsopgave

1 Root of Unity **2**
1.1 Root of Unity Filter . . . 2
1.2 Evaluating Sums . . . 2

2 Inequalities **2**
2.1 Standard Inequalities . . . 2
2.1.1 Schur's . . . 2
2.1.2 Cauchy-Schwarz . . . 2
2.1.3 AM-GM-HM-QM . . . 2
2.1.4 Weighted AM-GM . . . 2
2.1.5 Holder's . . . 2
2.1.6 Power Mean . . . 2
2.2 Nonstandard Inequalities . . . 2
2.3 Smoothing . . . 3
2.4 Lagrange Multipliers . . . 3

3 Polynomials **3**
3.1 Asymptotic Behavior . . . 3
3.2 Linear Constraints . . . 3
3.3 Interpolation . . . 3

4 Misc **3**
4.1 Finite Differences . . . 3
4.2 Fourier Transformations . . . 3

5 Analysis **4**

6 Algebraic Geometry **5**

1 Root of Unity

1.1 Root of Unity Filter

1.2 Evaluating Sums

2 Inequalities

2.1 Standard Inequalities

2.1.1 Schur's

2.1.2 Cauchy-Schwarz

2.1.3 AM-GM-HM-QM

2.1.4 Weighted AM-GM

2.1.5 Holder's

2.1.6 Power Mean

2.2 Nonstandard Inequalities

Theorem

Holder's Inequality For complex-valued functions f and g and reals p, q satisfying $\frac{1}{p} + \frac{1}{q} = 1$,

$$\|f^p\|^{\frac{1}{p}} \|g^q\|^{\frac{1}{q}} \geq \|fg\|$$

Example

2.3 Smoothing

2.4 Lagrange Multipliers

3 Polynomials

3.1 Asymptotic Behavior

3.2 Linear Constraints

3.3 Interpolation

4 Misc

4.1 Finite Differences

4.2 Fourier Transformations

Definition

Discrete Fourier Decomposition The decomposition is a systematic breakdown of a function $f(x)$ into sin and cos via

$$f(x) = \sum_{k=0}^{n} a_k \sin(kx) + \sum_{k=0}^{n} b_k \cos(kx)$$

and

$$a_k = \frac{2}{L} \int_0^L f(x) \sin(2\pi nkx) \mathrm{d}x$$

$$b_k = \frac{2}{L} \int_0^L f(x) \cos(2\pi nkx) \mathrm{d}x$$

Definition

Continuous Fourier Decomposition The continuous decomposition of a function $f(x)$ is

$$[x^d] f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-id\theta} f(e^{i\theta}) \mathrm{d}\theta$$

Problem

2025 CTST 15 Let X be a finite set of real numbers, d be a real number, and $\lambda_1, \lambda_2, \cdots, \lambda_{2025}$ be 2025 non-zero real numbers. Define

$$A = \left\{ (x_1, x_2, \cdots, x_{2025}) : x_1, x_2, \cdots, x_{2025} \in X \text{ and } \sum_{i=1}^{2025} \lambda_i x_i = d \right\},$$

$$B = \left\{ (x_1, x_2, \cdots, x_{2024}) : x_1, x_2, \cdots, x_{2024} \in X \text{ and } \sum_{i=1}^{2024} (-1)^i x_i = 0 \right\},$$

$$C = \left\{ (x_1, x_2, \cdots, x_{2026}) : x_1, x_2, \cdots, x_{2026} \in X \text{ and } \sum_{i=1}^{2026} (-1)^i x_i = 0 \right\}.$$

Show that $|A|^2 \leq |B| \cdot |C|$.

Solution

Define $f(t) = \sum_{x \in X} t^x$ to be a generating function. Then

$$|A| = \lim_{T \to \infty} \frac{1}{T} \int_0^T e^{-id\theta} \prod_{j=1}^{2025} f(e^{i\lambda_j \theta}) \mathrm{d}\theta$$

$$|B| = \lim_{T \to \infty} \frac{1}{T} \int_0^T f(e^{-i\theta})^{2024} \mathrm{d}\theta$$

$$|C| = \lim_{T \to \infty} \frac{1}{T} \int_0^T f(e^{-i\theta})^{2026} \mathrm{d}\theta$$

5 Analysis

Theorem

Gauss-Lucas Given a polnyomial $f(z) \in \mathbb{C}[x]$ with roots $z_1, z_2, \ldots, z_n$, its derivative $f'(z)$ with roots $w_1, w_2, \ldots, w_{n-1}$ lie within the convex hull of $z_1, \ldots, z_n$

Definition

Point Derivative The point derivative of a complex-valued function $f(z)$ is

$$D_\zeta f(z) = (\zeta - z)f'(z) + nf(z)$$

where $\deg(f) = n$.

Proposition

Gauss-Lucas for Point Derivative Let the roots of $D_\zeta f(z)$ be $w_1, \ldots, w_{n-1}$ and $f(z)$ be $z_1, \ldots, z_n$. Then $\max(\|w_1\|, \ldots, \|w_{n-1}\|) \leq \max(\|z_1\|, \ldots, \|z_n\|)$.

6 Algebraic Geometry